



Introduction

Bridging the Skills Gap

- Employer feedback consistently reveals that engineering graduates, while proficient in software and theoretical concepts, often lack direct experience with industrial equipment, manufacturing processes, and collaborative problem-solving. [1,2]
- This gap motivates the need for structured, hands-on experiential learning that mirrors real engineering workflows and prepares students for today's technology-driven workforce.

Motivation & Background

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- Real-world project engagement builds technical proficiency alongside teamwork, communication, and adaptability [4]. This workshop mirrors industry workflows from design through assembly using tools students will encounter in their careers

Workshop Overview

A two-part, hands-on workshop for undergraduate engineering students:

- PCB design and fabrication using Autodesk Fusion 360
- 3D-printed enclosure design using SolidWorks
- Component soldering and product assembly
- ESP8266 microcontroller programming (Arduino/C++)

Pre-Workshop Survey Results

Baseline confidence scores (1=No Experience, 5=Expert):

Skill Area	Mean	SD
CAD Software (SolidWorks / Fusion 360)	2.86	0.90
Electronics / Circuit Design	2.14	0.90
Manual Assembly / Soldering Tools	2.71	1.38
Soldering Iron Safety Confidence	3.86	1.68

Activity 1: PCB Design

Students follow the full professional PCB design workflow:

- Create component libraries from manufacturer datasheets (snapeda.com, ultralibrarian.com)
- Draw circuit schematics in Fusion 360, placing and connecting all components
- Design board layout with manual copper trace routing, meeting design rule requirements
- Export Gerber files for professional PCB manufacturing

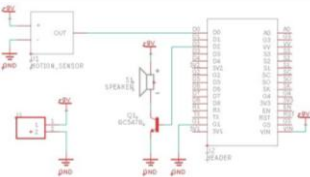


Figure 2: PCB Schematic



Figure 3: PCB Board Layout

Activity 2: 3D Enclosure Design

Using SolidWorks to design a custom enclosure for all components:

- Measure PCB, 9V battery, and ESP8266 to determine enclosure dimensions
- Add standoffs and slots to secure all internal components during use
- Include openings for connectors, switches, and ventilation
- Incorporate snap fits or screws; 3D-print in Library Makerspace

Activities 3 & 4: Assembly & Programming

Product Assembly:

Solder components (ESP8266 headers, transistors, switches, sensors) onto the PCB. Fit the PCB, battery, and module into the 3D enclosure per mounting features, then close and secure.

Microcontroller Programming:

Customize C/C++ code via Arduino framework to initialize variables, establish internet connectivity, continuously monitor sensor output, and send real-time mobile notifications upon alarm triggers.

Completed Product



Post-Workshop Survey Results

Post-workshop confidence scores (1=Not Confident, 5=Very Confident):

Skill Area	Mean	SD
3D Enclosure Design in SolidWorks	3.86	0.90
PCB Trace Routing (design rules)	3.86	0.90
Soldering (no bridges/cold joints)	4.29	0.95
Clarity of Technical Demonstrations	4.00	0.58

Conclusion & Future Work

- Students reported greater confidence, satisfaction, and readiness for future capstone and personal projects, while also expressing strong enthusiasm and commitment to their major.
- Responses related to sense of belonging were more mixed, highlighting opportunities to strengthen community connections. Future work will evaluate product quality as an additional measure of educational impact

References

- [1] Sheppard et al. Educating Engineers. 2009.
- [2] Pistrucci et al. Closing the Skills Gap. Automation Alley, 2021.
- [3] Finelli et al. Student Teams in the Engineering Classroom. CRLT, 2011.
- [4] Coyle et al. EPICS: Engineering Projects in Community Service. IJEE, 2005.